

◦ Aim:

- To determine the wavelength of sodium light using a transparent diffraction grating.

◦ Apparatus:

- spectrometer, diffraction grating, sodium lamp.

◦ Theory:

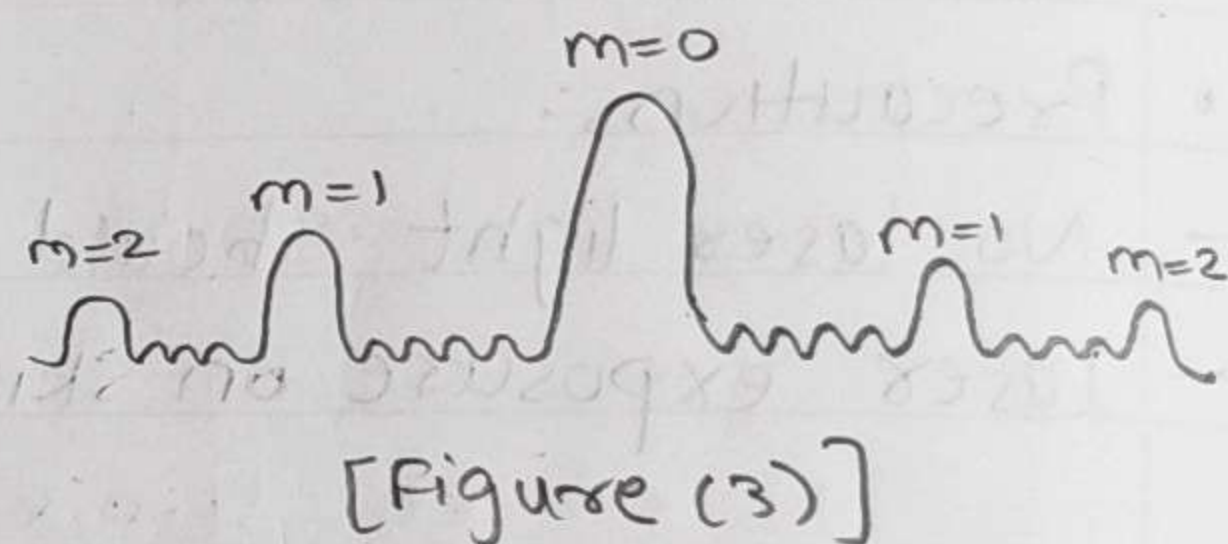
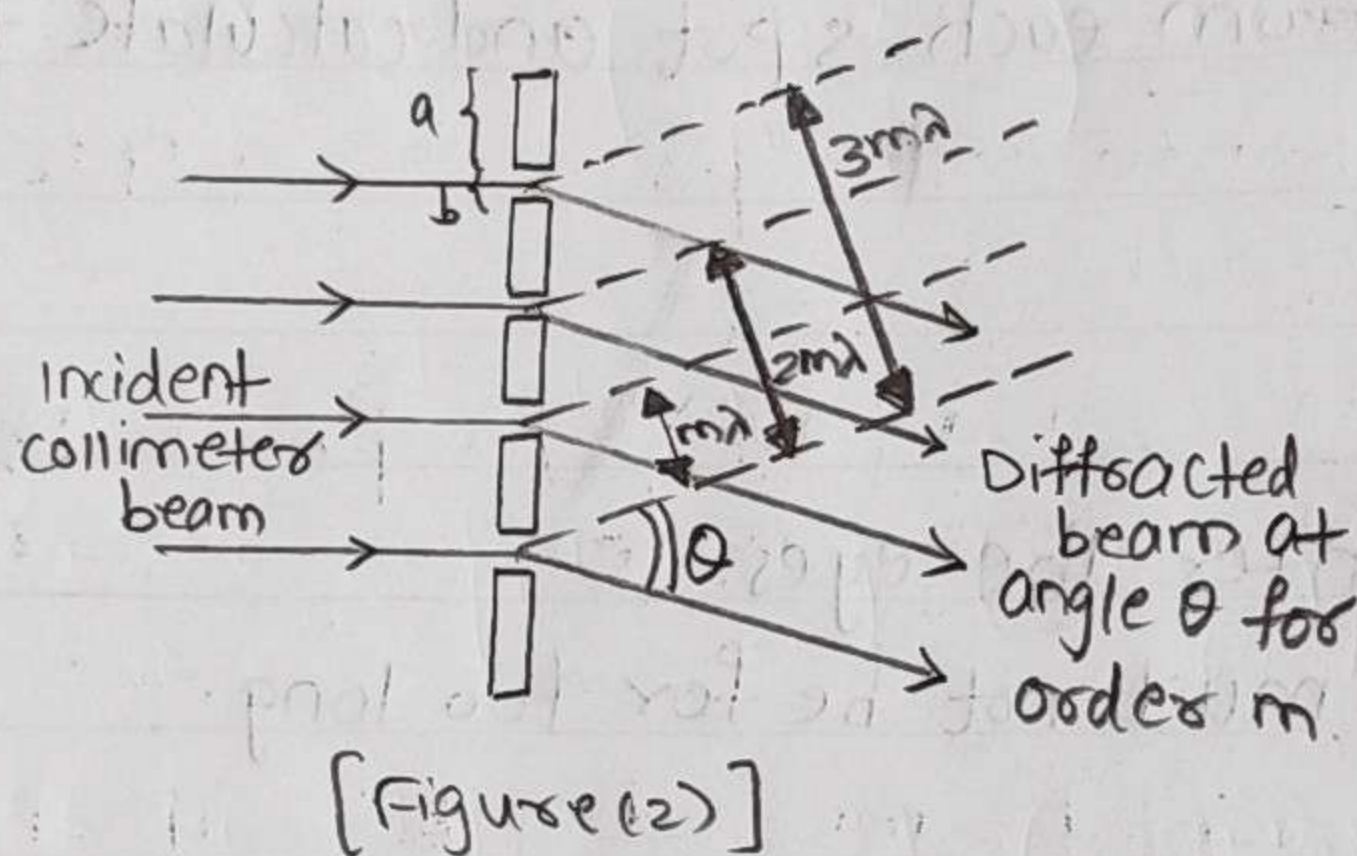
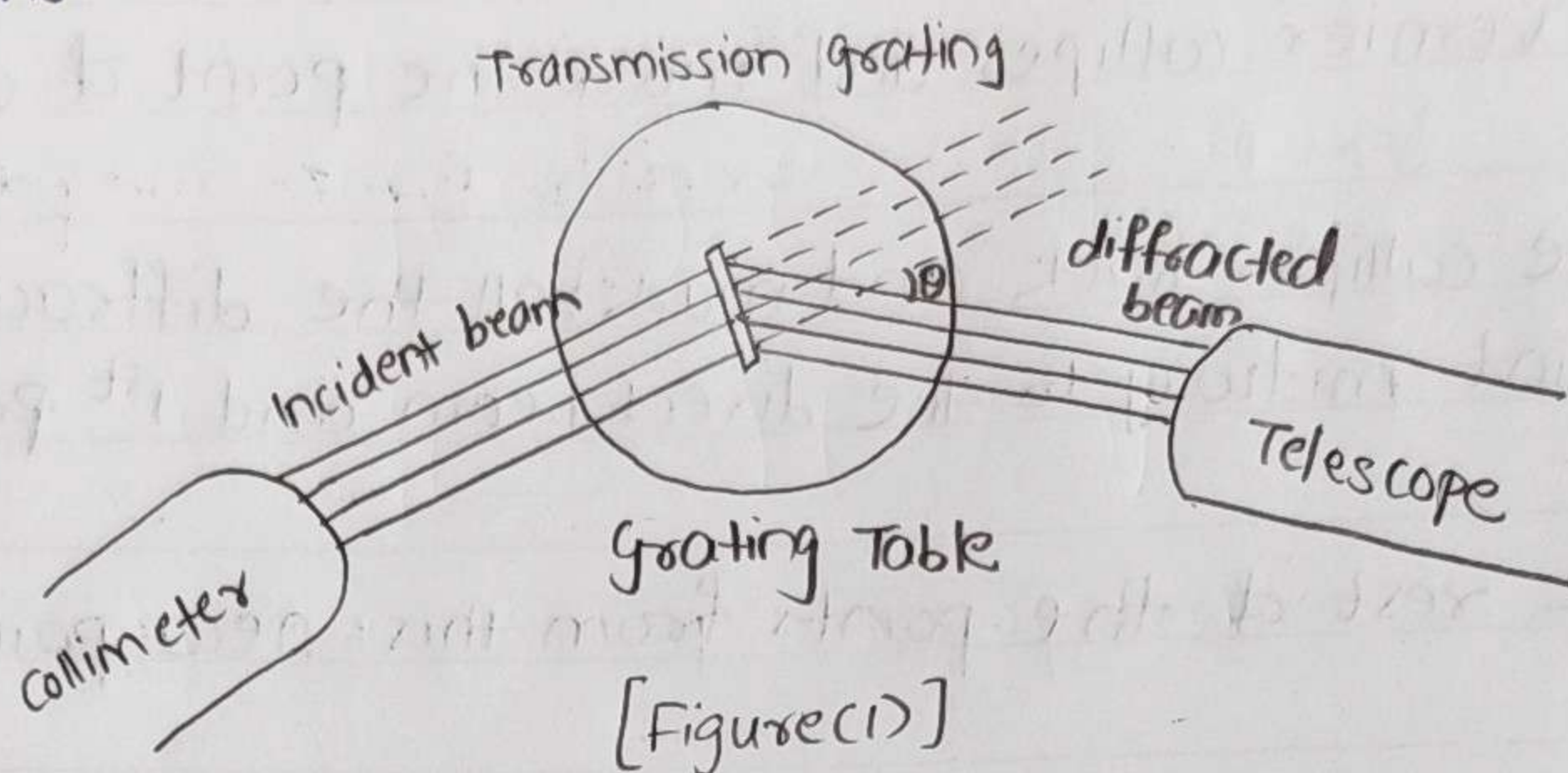
- The bending of light about sharp edges of an opaque obstacle or a narrow slit is called as 'diffraction'. Diffraction can be classified into two types:

(i) Fresnel diffraction: In this type of diffraction, the source and the screen are at a finite distance from the slit. Spherical or cylindrical wavefronts are formed here.

(ii) Fraunhofer Diffraction: In this type of diffraction, the source and screen are kept infinitely away from the slit. Planar wavefronts are formed here.

- Planar Transmission Diffraction Grating: It is the most simple & useful instrument for studying spectra. It consists of a grid of fine parallel lines uniformly spaced on a transmitting surface. The schematic diagram of a spectrometer is shown in Figure (1).

Diagrams:



Observation Table:

	Vernier V_1			Vernier V_2			Average
	Right θ_R	Left θ_L	$\theta_{V_1} = \frac{\theta_R - \theta_L}{2}$	Right θ_R	Left θ_L	$\theta_{V_2} = \frac{\theta_R - \theta_L}{2}$	
Green	$193^\circ 0'$	$159^\circ 2'$	16.983°	$352^\circ 25'$	$329^\circ 2'$	23.383°	20.1831°
Yellow	$194^\circ 4'$	$159^\circ 8'$	17.467°	$353^\circ 28'$	$327^\circ 15'$	26.216°	21.841°

o Calculations:

(i) for Green (1st order):

$$\theta_{v1} = \frac{\theta_R - \theta_L}{2} = \frac{1}{2} \left[\left(193 + \frac{0}{60} \right) - \left(159 + \frac{2}{60} \right) \right] = 16.983^\circ$$

$$\theta_{v2} = \frac{\theta_R - \theta_L}{2} = \frac{1}{2} \left[\left(352 + \frac{25}{60} \right) - \left(329 + \frac{2}{60} \right) \right] = 23.383^\circ$$

$$\therefore \theta_g = \frac{\theta_{v1} + \theta_{v2}}{2} = \frac{16.983 + 23.383}{2} = 20.1831^\circ$$

$$\therefore \lambda_g = (a+b) \sin \theta_g = \left(\frac{2.54 \times 10^{-2}}{15000} \right) \sin(20.1831^\circ) = 5.842 \times 10^{-7} \text{ m} = 5842 \text{ \AA}$$

(ii) for Yellow (2nd order):

$$\theta_{v1} = \frac{\theta_R - \theta_L}{2} = \frac{1}{2} \left[\left(194 + \frac{4}{60} \right) - \left(159 + \frac{8}{60} \right) \right] = 17.467^\circ$$

$$\theta_{v2} = \frac{\theta_R - \theta_L}{2} = \frac{1}{2} \left[\left(353 + \frac{28}{60} \right) - \left(327 + \frac{15}{60} \right) \right] = 26.216^\circ$$

$$\therefore \theta_y = \frac{\theta_{v1} + \theta_{v2}}{2} = \frac{17.467 + 26.216}{2} = 21.841^\circ$$

$$\therefore \lambda_y = (a+b) \sin \theta_y = \left(\frac{2.54 \times 10^{-2}}{15000} \right) \sin(21.841^\circ) = 6.299 \times 10^{-7} \text{ m} = 6299 \text{ \AA}$$

$$(iii) \lambda_{avg} = \frac{\lambda_g + \lambda_y}{2} = \frac{5842 + 6299}{2} = 6070.5^\circ$$

$$\therefore \% \text{ error} = \left| \frac{\lambda_{th} - \lambda_{avg}}{\lambda_{th}} \right| \times 100 = \left| \frac{6070.5 - 5893}{5893} \right| \times 100 = 3.01\%$$

- Assume 'a' be slit width and 'b' be distance between two adjacent slits, then $(a+b)$ is called as the grating element. If 'n' lines are ruled per unit length, then $(a+b) = 1/n$.
- The collimeter directs parallel light onto the grating and separates bundles of parallel light of different wavelengths and directs into the telescope; the angle θ is unique for each wavelength.
- The diffraction angle θ , wavelength λ , grating element $(a+b)$, and order 'm' are related as:

$$m\lambda = (a+b)(\sin i + \sin \theta) \quad \text{--- (i)}$$
- As incident light is perpendicular to plane of grating, $i = 0$.

$$\therefore m\lambda = (a+b) \sin \theta \quad \text{--- (ii)}$$
- From Figure (2), we can say if a grating element has M slits, then between any two principal maxima, there are $(M-1)$ secondary minima given by condition: $(a+b) \sin \theta = \pm \frac{n}{M} \lambda$
where $n \in \mathbb{Z} - \{0, M, 2M, \dots, kM, k \in \mathbb{Z}\}$.
as when $n = kM$ ($k \in \mathbb{Z}$), principal maxima is obtained.
- As in between two consecutive minima, there is a maxima, thus we can say that in between $(M-1)$ minima, there will be $(M-2)$ secondary maxima. The pattern obtained on screen is shown in Figure (3).

◦ Procedure:

- Adjust the collimeter and grating element such that parallel rays are incident onto the grating.
- Move the telescope and look for the first order image and take readings from both vernier scales towards the left side.
- Now, move telescope towards the right side and repeat above step.
- Subtract both the readings (θ_L and θ_R) and divide by 2 to get the angle from the first vernier (θ_{v1}). Repeat for second vernier (θ_{v2}).
- Take the average to get the angle of first order image (θ_1) and find the wavelength using given formula (λ_1).
- Repeat the above steps for second order image (λ_2).
- Take the average of both wavelengths and calculate the % error of calculate wavelength from actual wavelength of 5893 Å.

◦ Precautions:

- The experiment must be performed in a dark room.
- Do not touch the lines of the grating element.

o ~~Can~~ Applications:

- Sodium light is used in industrial lighting.

o Result:

- Wavelength of sodium lamp using spectrometer is $6070.5 \text{ \AA}$

o Conclusion:

- The wavelength of sodium lamp is found out to be $6070.5 \text{ \AA}$ with an error of 3.01% from the actual value of $5893 \text{ \AA}$.